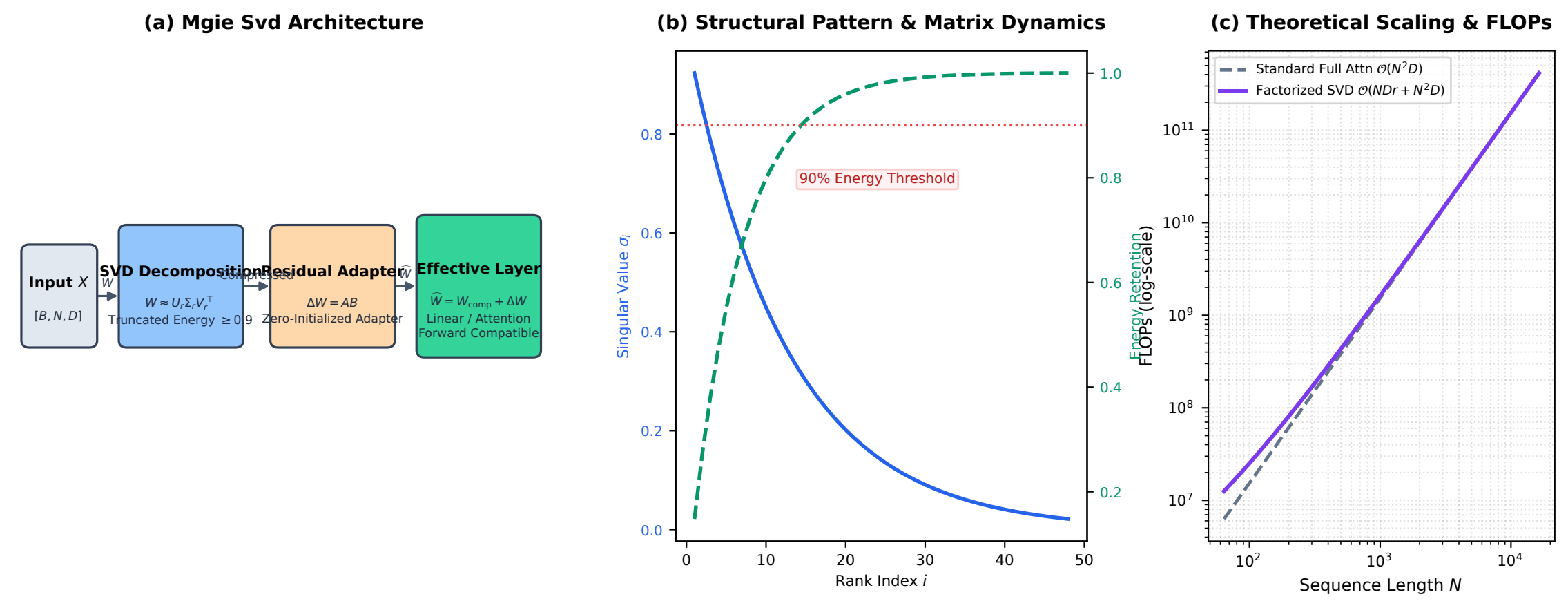
(a) Mgie Svd Architecture
Input $X$
$[B, N, D]$
$W$
SVD Decomposition
$W \approx U_r \Sigma_r V_r^\top$
Truncated Energy $\geq 0.9$
compressed
Residual Adapter
$\Delta W = AB$
Zero-Initialized Adapter
$\widehat{W}$
Effective Layer
$\widehat{W} = W_{\text{comp}} + \Delta W$
Linear / Attention
Forward Compatible
(b) Structural Pattern & Matrix Dynamics
90% Energy Threshold
Singular Value $\sigma_i$
Energy Retention
Rank Index $i$
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Factorized SVD $\mathcal{O}(NDr + N^2D)$
FLOPs (log-scale)
Sequence Length $N$